PPS Assignment

String & Array Problems in C

5 String Problems

5 Array Problems

Programming for Problem Solving • Solutions with Output

01

STRING PROBLEM

String Length

Take a string from the user and find its length without using the built-in strlen() function.

string-length.c

```c
#include <stdio.h>

int main(){

    char s[100];
    int i = 0;

    printf("Enter a string: ");
    scanf("%s", s);

    while(s[i] != '\0'){
        i++;
    }

    printf("Length = %d\n", i);

    return 0;
}
```

bash — output

```
$ ./string-length

Enter a string: programming
Length = 11
```

02

STRING PROBLEM

Reverse a String

Take a string and print all of its characters in reverse order.

reverse-string.c

```c
#include <stdio.h>

int main(){

    char s[100];
    int i = 0;

    printf("Enter a string: ");
    scanf("%s", s);

    while(s[i] != '\0'){
        i++;
    }

    printf("Reverse: ");
    for(int j = i-1; j >= 0; j--){
        printf("%c", s[j]);
    }
    printf("\n");

    return 0;
}
```

bash — output

```
$ ./reverse-string

Enter a string: hello
Reverse: olleh
```

03

STRING PROBLEM

Count Vowels & Consonants

Take a string and count how many vowels and how many consonants it contains.

count-vowels.c

```c
#include <stdio.h>

int main(){

    char s[100];
    int v = 0, c = 0;

    printf("Enter a string: ");
    scanf("%s", s);

    for(int i = 0; s[i] != '\0'; i++){

        char ch = s[i];

        if(ch=='a' || ch=='e' || ch=='i' || ch=='o' || ch=='u' || ch=='A' || ch=='E'
            v++;
        }else{
            c++;
        }
    }

    printf("Vowels = %d\n", v);
    printf("Consonants = %d\n", c);

    return 0;
}
```

bash — output

```
$ ./count-vowels

Enter a string: Education
Vowels = 5
Consonants = 4
```

04

STRING PROBLEM

Check Palindrome

Take a string and check whether it is a palindrome (reads the same forwards and backwards).

palindrome.c

```c
#include <stdio.h>

int main(){

    char s[100];
    int n = 0, ok = 1;

    printf("Enter a string: ");
    scanf("%s", s);

    while(s[n] != '\0'){
        n++;
    }

    for(int i = 0; i < n/2; i++){
        if(s[i] != s[n-1-i]){
            ok = 0;
            break;
        }
    }

    if(ok == 1){
        printf("Palindrome\n");
    }else{
        printf("Not palindrome\n");
    }

    return 0;
}
```

bash — output

```
$ ./palindrome

Enter a string: madam
Palindrome
```

05

STRING PROBLEM

Swap Letter Case

Take a string and flip the case: every small letter becomes capital and every capital becomes small.

change-case.c

```c
#include <stdio.h>

int main(){

    char s[100];

    printf("Enter a string: ");
    scanf("%s", s);

    for(int i = 0; s[i] != '\0'; i++){
        if(s[i] >= 'a' && s[i] <= 'z'){
            s[i] = s[i] - 32;
        }else if(s[i] >= 'A' && s[i] <= 'Z'){
            s[i] = s[i] + 32;
        }
    }

    printf("After change: %s\n", s);

    return 0;
}
```

bash — output

```
$ ./change-case

Enter a string: HelloWorld
After change: hELLOwORLD
```

06

ARRAY PROBLEM

Sum & Average

Take n numbers into an array and print their sum and their average.

array-sum.c

```c
#include <stdio.h>

int main(){

    int a[100], n, sum = 0;

    printf("How many numbers: ");
    scanf("%d", &n);

    for(int i = 0; i < n; i++){
        scanf("%d", &a[i]);
    }

    for(int i = 0; i < n; i++){
        sum = sum + a[i];
    }

    printf("Sum = %d\n", sum);
    printf("Average = %.2f\n", (float)sum/n);

    return 0;
}
```

bash — output

```
$ ./array-sum

How many numbers: 5
10 20 30 40 50
Sum = 150
Average = 30.00
```

07

ARRAY PROBLEM

Largest & Smallest

Take n numbers and find the biggest and the smallest value in the array.

max-min.c

```c
#include <stdio.h>

int main(){

    int a[100], n;

    printf("How many numbers: ");
    scanf("%d", &n);

    for(int i = 0; i < n; i++){
        scanf("%d", &a[i]);
    }

    int max = a[0];
    int min = a[0];

    for(int i = 1; i < n; i++){
        if(a[i] > max){
            max = a[i];
        }
        if(a[i] < min){
            min = a[i];
        }
    }

    printf("Max = %d\n", max);
    printf("Min = %d\n", min);

    return 0;
}
```

bash — output

```
$ ./max-min

How many numbers: 5
34 12 88 5 47
Max = 88
Min = 5
```

08

ARRAY PROBLEM

Reverse an Array

Take n numbers and print the whole array in reverse order.

reverse-array.c

```c
#include <stdio.h>

int main(){

    int a[100], n;

    printf("How many numbers: ");
    scanf("%d", &n);

    for(int i = 0; i < n; i++){
        scanf("%d", &a[i]);
    }

    printf("Reverse: ");
    for(int i = n-1; i >= 0; i--){
        printf("%d ", a[i]);
    }
    printf("\n");

    return 0;
}
```

bash — output

```
$ ./reverse-array

How many numbers: 5
1 2 3 4 5
Reverse: 5 4 3 2 1
```

09

ARRAY PROBLEM

Count Even & Odd

Take n numbers and count how many of them are even and how many are odd.

even-odd.c

```c
#include <stdio.h>

int main(){

    int a[100], n, even = 0, odd = 0;

    printf("How many numbers: ");
    scanf("%d", &n);

    for(int i = 0; i < n; i++){
        scanf("%d", &a[i]);
    }

    for(int i = 0; i < n; i++){
        if(a[i] % 2 == 0){
            even++;
        }else{
            odd++;
        }
    }

    printf("Even = %d\n", even);
    printf("Odd = %d\n", odd);

    return 0;
}
```

bash — output

```
$ ./even-odd

How many numbers: 6
11 22 33 44 55 66
Even = 3
Odd = 3
```

10

ARRAY PROBLEM

Search a Number

Take n numbers, then search for a number and print its position, or say Not found.

search-number.c

```c
#include <stdio.h>

int main(){

    int a[100], n, x, pos = -1;

    printf("How many numbers: ");
    scanf("%d", &n);

    for(int i = 0; i < n; i++){
        scanf("%d", &a[i]);
    }

    printf("Which number to search: ");
    scanf("%d", &x);

    for(int i = 0; i < n; i++){
        if(a[i] == x){
            pos = i;
            break;
        }
    }

    if(pos == -1){
        printf("Not found\n");
    }else{
        printf("Found at position %d\n", pos+1);
    }

    return 0;
}
```

bash — output

```
$ ./search-number

How many numbers: 5
10 20 30 40 50
Which number to search: 30
Found at position 3
```